S5 Glossary

Terms specific to this article, plus carried-over NDT/RVC/EAP and companion-article terms used throughout

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Robin Langell[1]
[1] Langell Konsult AB, Vallentuna, Sweden
For correspondence: Robin Langell, hello@noeticdiffusion.com

S5.0 Terms new to this article

Term	Definition
"State determines carrier"	The framing (`articles/Rythmic_Variance_Control_in_Sleep_theory/plan/02_response.md`, S8) that different sleep stages recruit different endogenous rhythmic events (N2 spindles, N3 slow oscillations, REM theta), and that autonomic phase-locking, if it reflects a general property of the underlying gating/denoising mechanism, should be testable - with possibly different strength - across all three, using one identical statistical test (R2).
Carrier	One of the three endogenous, stage-specific rhythmic events tested here for autonomic phase-locking: the N2 sleep spindle, the N3 slow oscillation (SO), and the REM-theta epoch. "Cross-carrier" = the same test applied to more than one carrier.
SO trough / SO upstate	Two reference points within a detected N3 slow oscillation at which cardiac/respiratory phase is measured: the trough (most negative deflection) and the upstate (rising transition following the trough) - analogous to spindle onset/peak's two reference points in the companion article.
Non-event risk set (`non_event_risk_v1`)	NMD's purpose-built sample of N2-stage timepoints, drawn per subject (stratified by time-of-night quartile, excluding a margin around every real spindle event), each carrying its own cardiac/respiratory phase - used as the "control" population in Test A's case-control hazard-style test (distinct from a parametric uniform-circular null).
Case-control logistic regression (Test A)	`is_case ~ cos(phase) + sin(phase)`, cluster-robust by subject, comparing real spindle onsets (cases) against the non-event risk set (controls); tests whether onset phase differs from the real, non-uniform N2 exposure distribution rather than an assumed-uniform one.
Circular time-shift surrogate (Test B)	A null construction that shifts each subject's own continuous phase stream by a random lag and re-reads phase at the same real event times, preserving the exact non-sinusoidal waveform/autocorrelation of that subject's own signal - a different null-construction principle from both Test A and the subject-rotation permutation (Null 0).
Null 0 / Test A / Test B	Article-specific labels for, respectively: the subject-rotation permutation test (unchanged from the companion article, relabeled here for direct comparison), the case-control logistic regression, and the circular time-shift surrogate - the three independent null constructions compared in R3.
Intrinsic spindle strength (qe)	A per-event measure of how strong/well-formed a detected spindle is, operationalized three ways: log1p-transformed sigma-power z-score (primary), log1p-transformed amplitude, and untransformed sigma-power z-score (robustness).
SO-spindle coupling	Whether a detected spindle has a temporally-matched slow-oscillation partner (`so_partner_missing`$= 0$) and, if so, the strength of that coupling (`so_spindle_coupling_score`) and the SO's own amplitude (`so_amplitude`) - available for only 11.8% of all spindles and 25/190 (13.2%) of the geometry-valid subsample.
Intrinsic-strength × alignment interaction	The regression term testing whether `resp_peak_alignment`'s effect on spindle-locked geometry delta depends on the spindle's own intrinsic strength (qe) - logically distinct from either variable's own main effect, and from the companion article's main-effect-only alignment-moderation test.
SO-coupled descriptive check	A small ($N = 25$ events/15 subjects), explicitly non-confirmatory correlational check (no interaction model, no BH-FDR family) between geometry delta and SO-coupling measures, reported for completeness/future hypothesis generation, not as a test result.

Figure 1: Terms introduced or newly operationalized in this article.

S5.1 Carried-over NDT/RVC/EAP and companion-article terms

Term	Definition
Noetic Diffusion Theory (NDT)	The parent geometric framework for neural state dynamics within which RVC and EAP are formulated ((Langell, 2025)); not itself tested by this article.
Rhythmic Variance Control (RVC)	The Tier-1 NDT mechanism under which endogenous, rhythm-defined events (spindles, SOs, REM-theta bursts) are treated as candidate transient gating-precision-change windows.
Embodied Anchoring Principle (EAP)	The Tier-2 NDT extension proposing that peripheral bodily rhythms can additionally condition the same latent landscape RVC events act on; this article's cross-carrier sweep and qe-interaction test are both relevant to (but not individually dispositive of) this principle.
Onset-phase vs. peak-phase	Two distinct reference points within a detected spindle event (companion article, (Langell, 2026)): `onset_sec` vs. `peak_sec` (time of maximum amplitude) - this article's N2-spindle rows in R2 test both, reproducing the companion article's own numbers exactly.
Event-timing pathway	The hypothesis that body phase helps determine **when** an endogenous RVC event (or a specific feature of it) occurs - operationalized here as non-uniformity of event phase relative to the cardiac/respiratory cycle, tested across all five carrier/reference-point conditions (R2-R3).
Event-efficacy pathway	The logically separate hypothesis that, given an event has occurred, the body's phase at that moment changes **how strongly** the event reconfigures state-space geometry - the companion article's alignment-moderation main-effect test, sharpened here into an interaction test with an intrinsic-strength covariate (R4).
Subject-rotation permutation null	A Monte Carlo null distribution (unchanged from the companion article) constructed by adding one i.i.d. $\mathrm{Uniform}(0, 2\pi)$ rotation per subject to all of that subject's phases, repeated n_{perm} times; destroys any common population-level preferred phase while exactly preserving each subject's event count and within-subject clustering.
Harmonic statistic (R_1, R_2, T_2)	R_1: first-harmonic mean resultant length (classical Rayleigh concentration). R_2: second-harmonic mean resultant length (sensitive to axial/bimodal structure). $T_2 = R_1^2 + R_2^2$: an omnibus statistic combining both, used throughout this article's and the companion article's permutation tests.
Held-out estimation	Estimating a population parameter (here, $\mu_{\mathrm{resp,peak}}$, recomputed self-containedly in this article, `S1.4`) on a data subset that excludes the events subsequently used to test a hypothesis involving that parameter, to avoid circularity/data leakage.
(Peak-phase) alignment	$\mathrm{resp_peak_alignment}_e = \cos(\varphi_e - \mu)$ for a given event e's respiratory-peak phase φ_e and the held-out population-preferred phase μ - unchanged from the companion article, used here as one of two interacting terms rather than as a standalone main effect.
WLS-cluster (`fit_wls_cluster`)	Weighted least squares with weights $1/n_{\mathrm{events,subject}}$ (equalizing each subject's total contribution regardless of event count) and cluster-robust standard errors by subject; the shared regression machinery for the qe-interaction model, inherited unchanged from both companion articles.
`d`, `d_n`, `d_l`, `d_s`, `e`, `e_e`, `e_s`, `e_m`, `m`	The 9 primary geometry endpoints inherited from the companion articles' `coords_9d` state-space representation; not redefined here.
`event_control` vs. `event_prefar` delta mode	Two ways of computing a geometry "delta" for a given event, inherited unchanged from the companion articles: `event_control` = event value minus a matched control window's value (a true event-vs-baseline contrast); `event_prefar` = event value minus the same event's own pre-spindle baseline ("fat") window (absolute change from pre-event).
BH-FDR	Benjamini-Hochberg false discovery rate correction, applied within each test family (10-test cross-carrier family, `S1.3`; 18-test qe-interaction family, applied separately per qe operationalization, `S1.5`) but not across families.

Figure 1: Terms inherited from the companion RVC/EAP article (`PAC_SLEEP_peak` analysis) and the broader NDT framework, used or tested in this article.

Bibliography

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